

Heart Disease Dataset

Exploratory Data Understanding & Discussion Report

1. Purpose

This report summarizes the cleaned dataset used for the heart disease analysis project, and works through the project questions the group defined, using descriptive statistics and cross-tabulations. It is intended as preparation material for the discussion session with the instructors: it explains what the data contains, what each question's answer looks like in numbers, and what insight or caveat is worth raising in discussion.

2. Dataset Overview

The cleaned dataset (project_data_cleaned.xlsx) contains 1,025 patient records across 20 columns, with no missing values remaining after cleaning. It combines raw clinical fields with grouped/binned versions created during cleaning (age group, blood pressure category, cholesterol category), which are the fields best suited for pivot table rows and columns.

Column summary

Column	Type	Description
Patient_ID	ID	Unique record identifier
age / Group_age	Numeric / Category	Patient age, and binned age group
sex	Category	Male / Female
chest_pain_type	Category	Typical Angina, Atypical Angina, Non-Anginal Pain, Asymptomatic
resting_blood_pressure / Group_blood_pressure	Numeric / Category	Resting BP (mm Hg) and clinical BP category
cholesterol / cholesterol_group	Numeric / Category	Serum cholesterol (mg/dl) and category
fasting_blood_sugar	Category	Above or below 120 mg/dl
rest_ecg	Category	Resting ECG result
Max_heart_rate	Numeric	Maximum heart rate achieved
exercise_induced_angina	Category	Yes / No
oldpeak	Numeric	ST depression induced by exercise
slope	Category	Slope of the peak exercise ST segment
vessels_colored_by_flourosopy / Num_vessels_colored	Category / Numeric	Number of major vessels (0-4) colored by fluoroscopy
thalassemia	Category	Normal, Fixed Defect, Reversible Defect, Unknown
target / heart_disease_flag	Binary / Category	1 = has heart disease, 0 = does not (Yes/No flag version)

3. Answers to the Project Questions

Q: What percentage of patients have heart disease?

51.3% of patients (526 of 1,025) are labeled with heart disease (target = 1); 48.7% (499) are not. The dataset is close to perfectly balanced, which is favorable for both descriptive analysis and any later classification modeling.

Q: Does age affect the risk of heart disease?

Age group	No disease %	Has disease %
Under 40	26.3	73.7
40-49	33.8	66.2
50-59	51.2	48.8
60-69	63.3	36.7
70+	41.2	58.8

The relationship is not a simple 'older = higher risk' pattern. Disease rate is actually highest in the under-40 and 40-49 groups, and lowest in the 60-69 group, before rising again at 70+. A likely explanation worth raising in discussion: younger patients in a clinical dataset are usually only tested when they already show strong symptoms, which inflates their positive rate (referral/selection bias), while the general population risk of heart disease still increases with age.

Q: Are males more likely to have heart disease than females?

Sex	Count	No disease %	Has disease %
Female	312	27.6	72.4
Male	713	57.9	42.1

Counter to common expectation, women in this sample have a much higher disease rate (72.4%) than men (42.1%). This is a strong candidate for discussion: it may reflect who was referred for testing rather than true population risk, since men outnumber women roughly 2:1 in the sample. Worth noting this is a well-known feature of the underlying (Cleveland) heart disease dataset and not a data error.

Q: What is the relationship between blood pressure and heart disease?

Blood pressure category	No disease %	Has disease %
Normal (<120)	41.8	58.2
Elevated (120-129)	50.0	50.0
Stage 1 HTN (130-139)	41.0	59.0
Stage 2 HTN (140-179)	56.4	43.6
Hypertensive crisis (180+)	82.4	17.6

There is no clean upward trend. Patients in the most severe blood-pressure category (180+) actually show the lowest disease rate (17.6%), while Normal and Stage 1 patients show higher rates. A plausible reason to raise: patients with very high blood pressure are often already being treated/medicated by the time this measurement

was taken, which can mask the underlying relationship. This is a good example of correlation not implying causation.

Q: Does cholesterol level affect heart disease?

Cholesterol category	No disease %	Has disease %
Desirable (<200)	41.4	58.6
Borderline High (200-239)	43.7	56.3
High (240+)	54.4	45.6

Similarly counter-intuitive: patients with 'Desirable' cholesterol have the highest disease rate, and patients with 'High' cholesterol have the lowest. This reinforces the same discussion point as blood pressure — single clinical values measured at one point in time do not map cleanly onto diagnosis outcome in this sample.

Q: Does fasting blood sugar increase the risk?

Fasting blood sugar	No disease %	Has disease %
Lower than 120 mg/dl	47.8	52.2
Greater than 120 mg/dl	53.6	46.4

The difference is small (about 6 percentage points) and in the opposite direction from what is clinically expected. Fasting blood sugar appears to be a weak predictor in this dataset.

Q: Which features are the strongest predictors of heart disease?

Feature	Correlation with target	Direction
Max_heart_rate	+0.42	Higher max heart rate -> more likely to have disease
oldpeak	-0.44	Higher ST depression -> less likely (label) to have disease*
Num_vessels_colored	-0.38	More vessels colored -> less likely (label) to have disease*
age	-0.23	Weak
resting_blood_pressure	-0.14	Weak
cholesterol	-0.10	Very weak

Max_heart_rate, oldpeak, and Num_vessels_colored are the strongest linear predictors of the target label — all comfortably ahead of age, blood pressure, and cholesterol, which are commonly assumed to be the primary risk factors. Categorical fields tell a consistent story: exercise-induced angina (66.9% disease rate when absent vs. 20.6% when present) and chest pain type (Typical Angina patients have only 24.5% disease rate vs. 77-80% for the other three types) also separate the classes strongly. *Note: the negative sign for oldpeak and vessel count reflects how this particular dataset's target is coded — worth double-checking the coding direction with the instructors, since a higher oldpeak and more blocked vessels are clinically expected to indicate more disease, not less.

Q: How is heart disease distributed across different age groups?

See the age table above (Question 2). Distribution by count: most patients fall in the 50-59 and 40-49 age bands, with the sample thinning out at the extremes (under 40 and 70+). The disease rate is not proportional to age band size.

Q: What insights can be obtained from the dataset?

- The dataset is well balanced (51% vs 49%), suitable for straightforward comparison and modeling without needing to correct for class imbalance.
- Traditional 'textbook' risk factors (age, blood pressure, cholesterol) show weak or inverted relationships with the diagnosis label in this sample, most likely due to referral/selection bias and pre-diagnosis treatment effects rather than the true clinical relationship.
- Exercise-test-based features (max heart rate, oldpeak, exercise-induced angina, chest pain type, number of vessels colored by fluoroscopy) are far stronger separators of the two classes than the baseline demographic/lab values.
- Women in the sample have a notably higher disease rate than men, which is a known characteristic of this dataset and worth flagging rather than treating as a general finding about sex-based risk.
- These patterns make a strong case for the group's chosen visualizations (box plots and violin plots especially) since they will make the age/BP/cholesterol counter-intuitive distributions visually obvious to the audience.

4. Supporting Statistics

Numeric field summary

Field	Mean	Std Dev	Min	Median	Max
age	54.43	9.07	29	56	77
resting_blood_pressure	131.61	17.52	94	130	200
cholesterol	246.00	51.59	126	240	564
Max_heart_rate	149.11	23.01	71	152	202
oldpeak	1.07	1.18	0.0	0.8	6.2

Max heart rate and oldpeak by diagnosis

Group	Mean Max_heart_rate	Mean oldpeak
No disease (target=0)	139.13	1.60
Has disease (target=1)	158.59	0.57

Patients with heart disease in this sample reach a notably higher average max heart rate during testing (158.6 vs 139.1 bpm) and show a lower average oldpeak (0.57 vs 1.60). These two fields, along with chest pain type and exercise-induced angina, are the best candidates to lead with in the group's presentation.